

**Exercise Sheet #7: Phasor Representation**

Discussion: Dec 1, 2025

**Exercise 1: Sinusoidal Voltage**Given the sinusoidal voltage  $v(t) = 50 \cos(30t + 10^\circ)$  V, find:

- a) the amplitude  $V_m$
- b) the period  $T$
- c) the frequency  $f$
- d)  $v(t)$  at  $t = 10$  ms.

**Solution**

- a) Amplitude:

$$V_m = 50 \text{ V}$$

- b) Period:

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{30} \approx 209.4 \text{ ms}$$

- c) Frequency:

$$f = \frac{1}{T} = \frac{30}{2\pi} \approx 4.775 \text{ Hz}$$

- d) Voltage at
- $t = 10$
- ms:

$$v(0.01) = 50 \cos(30 \cdot 0.01 + 10^\circ)$$

Convert  $10^\circ$  to radians:

$$10^\circ = \frac{10\pi}{180} = 0.17453 \text{ rad}$$

Compute the angle:

$$30 \cdot 0.01 + 0.17453 = 0.47453 \text{ rad}$$

Thus:

$$v(0.01) = 50 \cos(0.47453) \approx 44.48 \text{ V}$$

**Exercise 2: Phase Shift**Given  $v_1 = 45 \sin(\omega t + 30^\circ)$  V and  $v_2 = 50 \cos(\omega t - 30^\circ)$  V, determine the phase angle between the two sinusoids and which one lags the other.**Solution**Convert  $v_2$  to sine form:

$$v_2 = 50 \cos(\omega t - 30^\circ) = 50 \sin(\omega t - 30^\circ + 90^\circ) = 50 \sin(\omega t + 60^\circ)$$

Thus the phase angles are

$$\phi_1 = 30^\circ, \quad \phi_2 = 60^\circ$$

Phase difference:

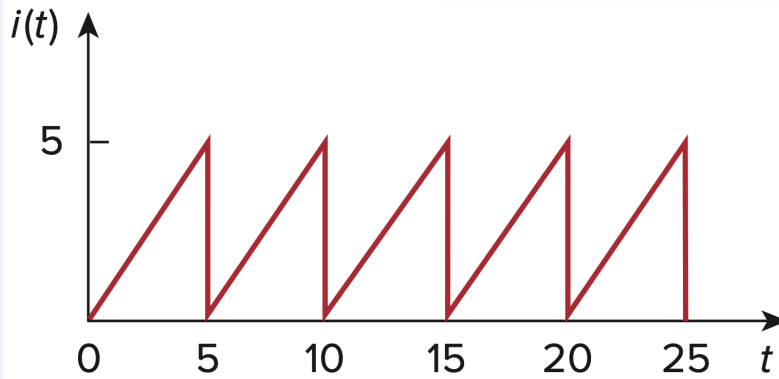
$$\Delta\phi = \phi_2 - \phi_1 = 60^\circ - 30^\circ = 30^\circ$$

Since  $\phi_1 < \phi_2$ ,

$v_1$  lags  $v_2$  by  $30^\circ$ .

### Exercise 3: RMS Value

Calculate the RMS value of the current waveform below.



### Solution

The waveform is a sawtooth with peak value 5 A and period  $T = 5$  s:

$$I = \sqrt{\frac{1}{T} \int_0^T i^2(t) dt} = \sqrt{\frac{1}{5} \int_0^5 \left(\frac{5t}{5}\right)^2 dt} = \sqrt{\frac{1}{5} \cdot \frac{1}{3} \cdot (5^3 - 0^3)} \text{ A} \approx 2.89 \text{ A}$$

### Exercise 4: Peak Phasors

For the following signals, find the complex amplitudes corresponding to a cosine reference as well as the rectangular form of the phasors.

- a)  $v(t) = 21 \cos(4t - 15^\circ) \text{ V}$
- b)  $i(t) = -8 \sin(10t + 70^\circ) \text{ mA}$

### Solution

- a) Peak phasor (cosine reference):

$$\underline{\hat{v}} = 21 \angle -15^\circ \text{ V}$$

Rectangular form:

$$\underline{\hat{v}} = 21 (\cos(-15^\circ) + j \sin(-15^\circ)) \text{ V} \approx (20.28 - j5.44) \text{ V}$$

- b) Convert to cosine form:

$$i(t) = -8 \sin(10t + 70^\circ) \text{ mA} = 8 \sin(10t + 250^\circ) \text{ mA} = 8 \cos(10t + 160^\circ) \text{ mA}$$

Peak phasor (cosine reference):

$$\hat{\underline{i}} = 8 \angle 160^\circ \text{ mA}$$

Rectangular form:

$$\hat{\underline{i}} = 8 (\cos(160^\circ) + j \sin(160^\circ)) \text{ mA} \approx (-7.52 + j2.74) \text{ mA}$$

### Exercise 5: Phasor Arithmetic

Two voltages  $v_1$  and  $v_2$  appear in series so that their sum is  $v = v_1 + v_2$ .  
If  $v_1 = 10 \cos(50t - \frac{\pi}{3}) \text{ V}$  and  $v_2 = 12 \cos(50t + 30^\circ) \text{ V}$ , find  $v$ .

### Solution

Convert both to phasors:

$$\hat{\underline{v}}_1 = 10 \angle -60^\circ \text{ V}, \quad \hat{\underline{v}}_2 = 12 \angle 30^\circ \text{ V}$$

Convert to rectangular form:

$$\hat{\underline{v}}_1 = 10 (\cos(-60^\circ) + j \sin(-60^\circ)) \text{ V} = (5 - j8.66) \text{ V}$$

$$\hat{\underline{v}}_2 = 12 (\cos(30^\circ) + j \sin(30^\circ)) \text{ V} = (10.39 + j6) \text{ V}$$

Sum the phasors:

$$\hat{\underline{v}} = \hat{\underline{v}}_1 + \hat{\underline{v}}_2 = ((5 + 10.39) + j(-8.66 + 6)) \text{ V} = (15.39 - j2.66) \text{ V}$$

Convert back to polar form:

$$|\hat{\underline{v}}| = \sqrt{15.39^2 + (-2.66)^2} \text{ V} \approx 15.62 \text{ V}$$

$$\theta = \tan^{-1} \left( \frac{-2.66}{15.39} \right) \approx -9.8^\circ$$

Thus:

$$\hat{\underline{v}} = 15.62 \angle -9.8^\circ \text{ V}$$

Convert back to time domain:

$$v(t) = 15.62 \cos(50t - 9.8^\circ) \text{ V}$$

### Exercise 6: Impedance

A series  $RLC$  circuit has  $R = 80 \Omega$ ,  $L = 240 \text{ mH}$ , and  $C = 5 \text{ mF}$ .  
If the input voltage is  $v(t) = 10 \cos(2t) \text{ V}$ , find the current flowing through the circuit.

### Solution

Angular frequency:

$$\omega = 2 \text{ rad/s}$$

Impedances:

$$\underline{Z}_R = 80 \Omega$$

$$\underline{Z}_L = j\omega L = (j2 \cdot 0.24) \Omega = j0.48 \Omega$$

$$\underline{Z}_C = \frac{1}{j\omega C} = \frac{1}{j2 \cdot 0.005} \Omega = -j100 \Omega$$

Total impedance:

$$\underline{Z} = \underline{Z}_R + \underline{Z}_L + \underline{Z}_C = 80 \Omega + j0.48 \Omega - j100 \Omega = (80 - j99.52) \Omega$$

Magnitude and phase of total impedance:

$$|\underline{Z}| = \sqrt{80^2 + (-99.52)^2} \Omega \approx 127.69 \Omega$$

$$\theta_Z = \tan^{-1} \left( \frac{-99.52}{80} \right) \approx -51.21^\circ$$

Phasor form of voltage:

$$\underline{\hat{v}} = 10 \angle 0^\circ \text{ V}$$

Phasor form of current:

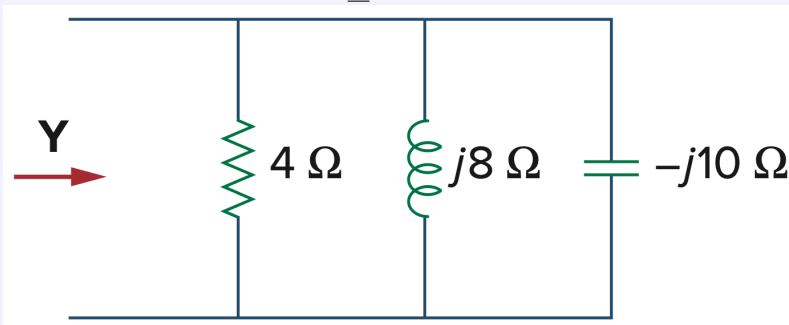
$$\underline{\hat{i}} = \frac{\underline{\hat{v}}}{\underline{Z}} = \frac{10 \angle 0^\circ \text{ V}}{127.69 \angle -51.21^\circ \Omega} = \frac{10 \text{ V}}{127.69 \exp(-j51.21^\circ) \Omega} \approx 78.3 \angle 51.21^\circ \text{ mA}$$

Convert back to time domain:

$$i(t) = 78.3 \cos(2t + 51.21^\circ) \text{ mA}$$

### Exercise 7: Admittance

Determine the admittance  $\underline{Y}$  for the circuit shown below.



### Solution

Calculate the individual admittances:

$$\underline{Y}_R = \frac{1}{\underline{Z}_R} = \frac{1}{4 \Omega} = 0.25 \text{ S}$$

$$\underline{Y}_L = \frac{1}{\underline{Z}_L} = \frac{1}{j8 \Omega} = -j0.125 \text{ S}$$

$$\underline{Y}_C = \frac{1}{\underline{Z}_C} = \frac{1}{-j10 \Omega} = j0.1 \text{ S}$$

Total admittance:

$$\underline{Y} = \underline{Y}_R + \underline{Y}_L + \underline{Y}_C = 0.25 \text{ S} - j0.125 \text{ S} + j0.1 \text{ S} = (250 - j25) \text{ mS}$$